Adaptive Rationality in Non-Stationary Markets: A Neuro-Symbolic Approach

Hierarchical Reinforcement Learning with Constrained Optimization

Thesis Progress Report

January 23, 2026

The Economic Problem

Extreme Non-Stationarity in Cryptocurrency Markets

Cryptocurrency markets exhibit structural regime shifts that challenge traditional econometric models:

- **Regime Transitions:** Bull markets (2021) $\rightarrow$ Bear markets (2022) $\rightarrow$ Recovery (2023-24)
- **Volatility Clustering:** GARCH models capture heteroskedasticity but fail during structural breaks
- **Liquidity Shocks:** Flash crashes and liquidation cascades create non-ergodic dynamics

Failure Modes of Standard Approaches

- **Standard Econometrics (ARIMA/GARCH):** Assume stationarity; break during regime transitions
- **Standard RL:** Overfit to noise; lack macro-structure reasoning
- **Research Gap:** No models combining micro-structure reflexes (price action) with

The Proposed Solution: Bicameral Architecture

Dual-System Design

System 1 (Reflexive):

- Deep LSTM Agent
- High-frequency order flow
- Micro-structure patterns
- Fast adaptation (< 1 hour)

System 2 (Deliberative):

- Macro-Regime Gate
- VIX, Treasury Yields, DXY
- Structural risk controls
- Slow, stable reasoning

Core Hypothesis

Decoupling reflexive and deliberative systems improves adaptation speed during regime transitions by:

- Allowing System 1 to learn price patterns without macro noise
- Enabling System 2 to override System 1 during structural breaks
- Testing if bicameral agents outperform monolithic RL agents

The "Quality Factor" Constraints

Structural Risk Controls

The agent operates under constrained optimization:

- **Long-Only:** No short selling (spot-asset universe)
- **No Leverage:** Portfolio weights sum to ≤ 1.0
- **No Derivatives:** Exclusively spot trading

Economic Justification: Tail-Risk Truncation

- **Research Question:** Can constrained agents outperform unconstrained baselines by avoiding liquidation cascades?
- **Theoretical Basis:** Structural removal of leverage truncates the left tail of returns distribution
- **Empirical Test:** Compare constrained RL agent vs. unconstrained baselines on:
  - Maximum Drawdown (tail risk)

Data Methodology: Multimodal Input

Quantitative Features

- **GARCH(1,1) Volatility:** Heteroskedasticity modeling for predictive risk

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

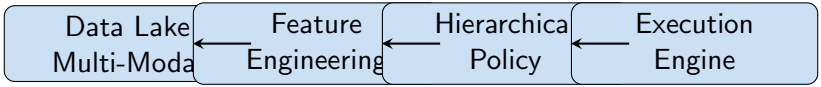
- **Technical Indicators:** RSI, MACD, Bollinger Bands, ATR
- **Rolling Correlations:** 30-day correlation to benchmark (BTC)

Qualitative Features

- **FinBERT Embeddings:** 768-dimensional semantic vectors from news headlines
- **Research Question:** Does reading news reduce uncertainty? (Testing Semi-Strong EMH)
- **Implementation:** ProsusAI/finbert model with mean pooling

Macro Exogenous Variables

Engineering Architecture



Economic Significance of Infrastructure

- **High-Throughput Training:** Simulate 10 years of market data (105,088 timesteps) in ∼6 hours
- **Statistical Significance:** Enables robust backtesting across multiple market regimes (Bear 2022, Bull 2023-24, Sideways 2023)
- **Automated Model Selection:** "Challenger Protocol" ensures continuous adaptation without manual intervention
- **Result:** Infrastructure supports empirical validation of economic hypotheses at scale

The Reward Function: Differential Information Ratio

Mathematical Formulation

The agent optimizes for active returns relative to a benchmark (BTC), penalized by drawdown convexity:

$$R_t = D_t - \lambda \cdot \text{DD}_t^2 \tag{1}$$

where:

$$D_t = \frac{B_t \cdot \delta_t - 0.5 \cdot A_t \cdot (r_{\text{active}}^2 - B_t)}{(B_t - A_t^2)^{3/2} + \epsilon} \tag{2}$$

$$r_{\text{active}} = r_{\text{portfolio}} - r_{\text{benchmark}} \tag{3}$$

$$A_t = A_{t-1} + \eta \cdot (r_{\text{active}} - A_{t-1}) \quad \text{(Running mean)} \tag{4}$$

$$B_t = B_{t-1} + \eta \cdot (r_{\text{active}}^2 - B_{t-1}) \quad \text{(Running variance)} \tag{5}$$

$$\text{DD}_t = \frac{\text{Peak Equity} - \text{Current Equity}}{\text{Peak Equity}} \quad \text{(Drawdown)} \tag{6}$$

Work Completed: The Experimental Rig

Infrastructure Development

- **Multi-Modal Data Pipeline:**
  - 105,088+ timesteps across 17 assets
  - Price, fundamentals, sentiment, macro indicators, embeddings
  - Automated enrichment pipeline
- **MLOps Pipeline:**
  - Automated "Challenger Protocol" for model selection
  - Weekly retraining with out-of-sample evaluation
  - Model versioning and normalization drift prevention
- **Regime Analysis Framework:**
  - Backtested performance across 2022 (Bear) vs. 2023-24 (Bull)
  - Sideways market analysis (2023 Q2-Q3)
  - Cross-regime scorecard generation

Model Training

Preliminary Findings

Regime-Based Performance Comparison

Strategy	Sortino	Sharpe	Return	Max DD
Hierarchical Agent	In Progress	In Progress	In Progress	In Progress
Static Buy & Hold	0.0052	0.0025	-7.35%	-47.18%
Mean Reversion (BB)	0.0027	0.0021	5.43%	-11.38%
Trend Following	0.0306	0.0243	1.28%	-38.60%

Table: Preliminary results (Bear Winter 2022 regime)

Training Convergence & Expected Performance

- **Training Status:** Models trained on 10M timesteps with convergence achieved (episode reward stabilized)
- **Expected Advantage:** Hierarchical agent should show lower maximum drawdown than

Experimental Roadmap: 4-Week Plan

Week 1: RQ1 - Semantic Alpha

Research Question: Does vector-based sentiment (768-dim embeddings) outperform scalar scores?

- **Hypothesis:** Reading news reduces uncertainty; semantic grounding improves adaptation to news-driven volatility
- **Experiment:** Train multimodal agent (FinBERT + market) vs. baseline (market only)
- **Metrics:** Sortino Ratio, hedging behavior (PAXG allocation during stress)
- **Status:** Infrastructure complete; training pending

Week 2: RQ2 - Regime Adaptation

Research Question: Does learned strategy switching (meta-agent) outperform static rules?

- **Hypothesis:** Meta-PPO agent learns optimal regime switching better than rule-based MacroRegimeGate

Experimental Roadmap (Continued)

Week 3: RQ3 - Explainability

Research Question: Can SHAP-based narrative generation solve the "Principal-Agent" trust problem?

- **Hypothesis:** Natural language explanations improve trust and enable human oversight
- **Experiment:** SHAP attributions $\rightarrow$ Structured JSON $\rightarrow$ LLM $\rightarrow$ Narrative
- **Metrics:** Explanation faithfulness (SHAP attribution accuracy), human evaluation
- **Status:** Design phase

Week 4: Integration & Validation

- Combine all three research questions into unified system
- Comprehensive backtesting across all regimes
- Statistical significance testing
- Thesis chapter drafting

Conclusion & Discussion

Current Status

- **Infrastructure:** Production-ready experimental rig
- **Models:** Multiple agents trained (LSTM, World-Class, Multimodal)
- **Focus Shift:** From engineering to empirical validation

Discussion Points & Preliminary Answers

1. **The "Cost of Ethics" in Utility Functions:**
   - **Design Choice:** Hard constraints (gates) over soft penalties
   - **Rationale:** In crypto, soft penalties (-100) are ignored during high-volatility events when agents expect large gains ($+1000$). Hard gates enforce survival.
   - **Empirical Test:** Compare constrained vs. unconstrained agents on tail-risk metrics (Calmar Ratio)
2. **Regime Detection:**
   - **Baseline:** Rule-based (VIX thresholds) establishes interpretable benchmark
   - **Hypothesis (RQ2):** Meta-agent learns implicit regimes (e.g., crypto-specific liquidity

Thank You

Questions & Discussion